6.6 Mechanism of evolution

Microevolution (evolution on a small scale) refers to the changes in allele frequencies within a population. Allele frequencies in a population may change due to four fundamental forces of evolution such as natural selection, genetic drift, mutation and gene flow.

6.6.1 Natural selection

It occurs when one allele (or combination of alleles of differences) makes an organism more or less fit to survive and reproduce in a given environment. If an allele reduces fitness, its frequencies tend to drop from one generation to the next.

The evolutionary path of a given gene i.e., how its allele's change in frequency in the population across generation, may result from several evolutionary mechanisms acting at once. For example, one gene's allele frequencies might be modified by both gene flow and genetic drift, for another gene, mutation may produce a new allele, that is favoured by natural selection **(Fig. 6.6).**

Selection

There are mainly three types of natural selection

i. **Stabilising Selection (centipetal selection):** This type of selection operates in a stable environment **(Fig. 6.7a).** The organisms with average phenotypes survive whereas the extreme individuals from both the ends are eliminated. There is no speciation but the phenotypic stability is maintained within the population over generation. For example, measurements of sparrows that survived the storm clustered around the mean, and the sparrows that failed to survive the storm clustered around the extremes of the variation showing stabilizing selection.

ii. **Directional Selection:** The environment which undergoes gradual change is subjected to directional selection **(Fig. 6.7b)**. This type of selection removes the individuals from one end towards the other end of phenotypic distribution. For example, size differences between male and female sparrows. Both male and female look alike externally but differ in body weight. Females show directional selection in relation to body weight.

iii. **Disruptive Selection (centrifugal selection):** When homogenous environment changes into heterogenous environment this type of selection is operational **(Fig. 6.7c).** The organisms of both the extreme phenotypes are selected whereas individuals with average phenotype are eliminated. This results in splitting of the population into sub population/species. This is a rare form of selection but leads to formation of two

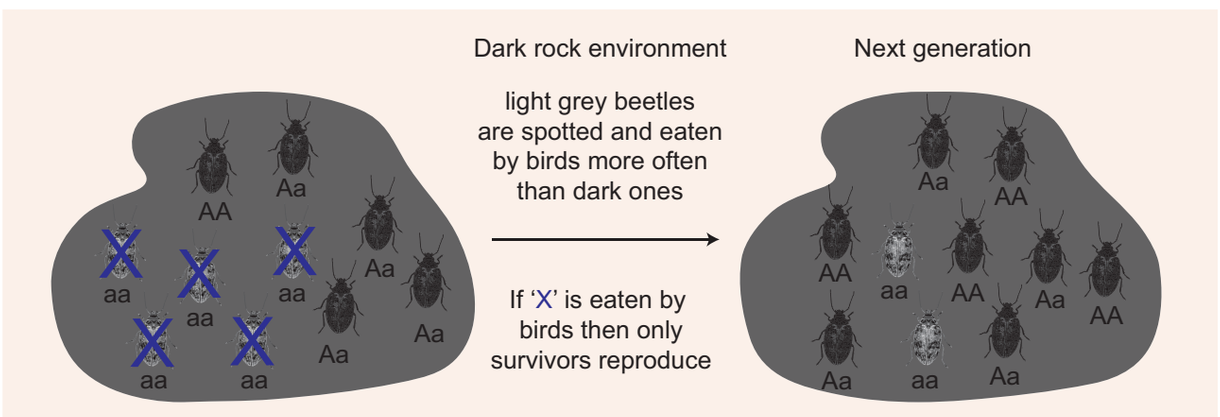


Fig 6.6 Natural selection

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